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**r/GooglePixel** • 5mo ago

HeWhoShantNotBeNamed Pixel 10 Pro

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The zoom image processing has changed on the 10 Pro

Every year, since the Pixel 7 Pro, I've done an in-depth analysis on Pixel image processing at various zoom levels. Here's my post from last year: <https://www.reddit.com/r/GooglePixel/s/OEbjlhStmU>

This year I tested it on text and what I noticed is that the telephoto overlay merged into the main image at zoom levels between 2.5x and 4.8x doesn't actually take the raw telephoto image, it seems to actually be fusing the images with an algorithm rather than simply overlaying the telephoto onto the main.

The 4.7x looks significantly worse than the 5x. Though it's still way better than anything below 2.5x.

Another think I noticed is overprocessing at 2x. At 2x, the camera takes the middle pixels rather than pixel binning to get a native 12 MP.

The problem is that text looks horrible. It looks worse than 1.9x. It looks much worse than 50 MP. Both of these are the opposite of what has been true in the past. Granted I'm testing on a different object than I have in the past (usually tested on grass and leaves), but at 2x I'm getting weird colored artifacting not present at 1.9x zoom on text. And it's less legible.

It's not the quality of the image, it's the over sharpening and aggressive denoising that is causing a loss in actual detail in text. That being said, zooming into 4.5x at 12 MP will still look better than a 50 MP shot at any zoom.

At this stage I'd rather use 50 MP up to 2.4x rather than switch to 12 MP like in previous models when going for clarity/resolution. Google needs to fix their "middle pixel" sensor processing for 2x - 2.4x photos.



121



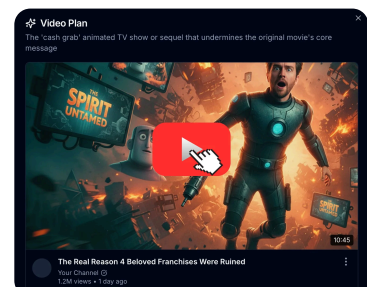
53

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**aerog16** • 5mo ago

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**HeWhoShantNotBeNamed** OP • 5mo ago

Pixel 10 Pro

It's basically the same except that 2x looks bad now and 50 MP no longer has shutter lag in most conditions.

**aerog16** • 5mo ago

What happened to make 2x worse than last year? It still takes the middle pixels to get to 12 MP and super res? Isn't that the same processing as last year?

**HeWhoShantNotBeNamed** OP • 5mo ago

Pixel 10 Pro

Yeah I'm not sure why it looks worse with text. I'm gonna test on my usual grass today when it gets brighter outside to see if maybe it's just text processing.

**dextroz** • 5mo ago

The reality is that you care about this more than any of the Google Pixel teams. None of their product managers are long-term and all they're interested in is a launch, grab that ensuing promotion, and peace out to the next Minimal Viable Effort gig in the company.

**HeWhoShantNotBeNamed** OP • 5mo ago

Pixel 10 Pro

**cardonator** • 5mo ago

Pixel 10 Pro XL

**JanCapek** • 5mo ago

Pixel 9 Pro

Thanks for technical explanation of [my findings](#).

**HeWhoShantNotBeNamed** OP • 5mo ago

Pixel 10 Pro

So it's not just me.

**cbelliott** • 5mo ago

Question for you OP... I don't know where to make this change but I saw a review recently on YouTube and they said that you have to go into the camera setting somewhere and there's an option that you can toggle to where you can force the phone to use the exact camera you are trying to use. So you can force it to use the main or the wide or the telephoto according to what you were trying to shoot. Do you think that this modification would change any of your results if you were able to force it to use the telephoto when you were trying to make a comparison of the quality of the telephoto compared to the other

[Skip to main content](#)[+](#) Create**HeWhoShantNotBeNamed** OP • 5mo ago

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Yes you can manually select the lens. I'm sure it would change some things. My guess is that it would function as normal in the intended zoom ranges.

The telephoto consistently switches at 4.8x unless it's super dark so I don't see a need to do this. I take a LOT of photos and video and never manually switch the lens, I don't see the point. There are very few instances where perhaps manually selecting the telephoto would be preferable since it'll avoid using it if it's super dark.

**[deleted]** • 5mo ago

Telephoto lens is switched at 5x, not at 4.8x

**HeWhoShantNotBeNamed** OP • 5mo ago

Pixel 10 Pro

No it is 4.8x. Try zooming really slowly.

[r/confidentlyincorrect](#)

**lssong99** • 4mo ago

On P10PXL my test confirmed at 4.8x

**pspr33** • 5mo ago

On my P9PXL it changes at 4.9x, just tested it. You have to go really slow with the zoom to see.

**StimulatorCam** • 5mo ago • Edited 5mo ago

Pixel 8 Pro Top 1% Commenter

None of your links are working.

Edit: They work now, thanks.

**HeWhoShantNotBeNamed** OP • 5mo ago

Pixel 10 Pro

They should work now. I hate that this subreddit has banned link shorteners.

**Blunt552** • 5mo ago

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but Google has proven me wrong, interestingly enough only on main sensor.

when looking at GSMarenas review the telephoto is much cleaner than the mess of the main sensor.

Did you also do some tests in low light on this gen or are you planning to?



HeWhoShantNotBeNamed OP • 5mo ago

Pixel 10 Pro

All my low light pictures have looked great so I'm not sure what you're talking about.



AppropriatePage273 • 5mo ago

<https://www.youtube.com/watch?v=cvwAXJh6sLg> Please check this video



HeWhoShantNotBeNamed OP • 5mo ago

Pixel 10 Pro

That video doesn't show low light photos "super processed and artificial"



AppropriatePage273 • 5mo ago

Check 6:10 It seems you just want to believe what you want to.



HeWhoShantNotBeNamed OP • 5mo ago

Pixel 10 Pro

The person in the video made no claim of any such thing at that timestamp nor did I see "artifacts and super processed" in any capacity.



Blunt552 • 5mo ago

Yeah, there is definitely something funky going on with the main sensor, it looks super artificial and grainy in low light, but not noise wise, I can't put my finger on it, it just looks strange.



Blunt552 • 5mo ago

Could you share some? The samples I've seen on GSMArena and DXOMark were horrible.



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future fine waiting tie terrific mysterious juggle dinosaurs versed vanish

This post was mass deleted and anonymized with [Redact](#)**HeWhoShantNotBeNamed** OP • 5mo ago

Pixel 10 Pro

You can use Video Boost on ultrawide which is nice.

**BrandonBoi** • 5mo ago

Pixel 9 Pro

I've noticed the 50MP shots are instantaneous on the main camera now as opposed to taking half a second or so to take. The delay is still there for the ultrawide and telephoto though but definitely a noticeable improvement. However, I've also noticed that because the shots are instantaneous now, they do a worse job at letting light in during lower light shots. This isn't necessarily a problem, just an interesting observation as the prior 50MP shots were almost like a pseudo night mode due to taking a bit to process

**StimulatorCam** • 5mo ago

Pixel 8 Pro Top 1% Commenter

I did some shots of my own with my 8 Pro from 1x-4.5x using both 12MP and 50MP but also in both auto and manual lens selection. I used a phone stand to keep every shot as equal as possible, but have a look to see what you think of my examples.

<https://imgur.com/a/jbbgQQG>**HeWhoShantNotBeNamed** OP • 5mo ago

Pixel 10 Pro

Imgur isn't great because of image compression, but even in your sample you can clearly see that manual lens selection turns off other processing and ruins the quality.

**StimulatorCam** • 5mo ago • Edited 5mo ago

Pixel 8 Pro Top 1% Commenter

Lol wut? Look at the details, especially in the 2.5x and 4.5x, there's lots of bizarre sharpening going on in the auto mode photos.

Edit: I mean yeah the 12MP manual is the worst of the 4 options, but the 12MP auto still falls far behind either 50MP option.

-  Create 
-  Pixel 10 Pro
I wouldn't call it "bizarre". 12 MP Auto looks great at 2.5x to me. As well as 4.5x.
-  **StimulatorCam** • 5mo ago
Pixel 8 Pro  Top 1% Commenter
Look at the green sections, the squiggly sharpening artefacts are the most noticeable there. And I'm not saying the 12MP auto looks overly bad, but in comparison to the 50MP shots it's lacking. But this shouldn't be happening at these digital zoom levels where the 50MP isn't even getting any more pixels to work with than the 12MP.
-  **HeWhoShantNotBeNamed** OP • 5mo ago
Pixel 10 Pro
I wonder if it's trying to use AI (not generative, just old fashioned AI) to guess what it actually should look like. Or maybe it's out of focus or blurry.
-  **StimulatorCam** • 5mo ago
Pixel 8 Pro  Top 1% Commenter
So I did more rounds of testing, but at a slightly further distance and the range of >2x and <5x definitely improved with the 12MP auto, because I think in my previous test the distance was too close for the telephoto lens to focus so it wasn't combining the two. At a further distance it has that option.

But I also tested the ultrawide and telephoto more thoroughly and have to agree with your previous analysis that 50MP makes essentially no difference with the ultrawide, but I found that with the telephoto the 50MP is always better than the 12MP even at 10x, 20x, 30x. Again this makes no sense as there's no extra pixels for the 50MP mode to use at those crop levels, but in my tests it consistently did noticeably better.
-  **Baspower** • 5mo ago
Pixel 10 Pro
I agree about the 10x. 12mp images come out very oversharpened for me, while 50mp images look more natural.